

Project Proposal

SkillForge AI: An Agentic AI Career Operating System

1. Project Overview

SkillForge AI is an intelligent, multi-agent career operating system designed to guide students and professionals throughout their complete learning and career journey. Unlike traditional AI chatbots that only answer questions, SkillForge AI acts as a personalized AI mentor, career coach, and automation assistant.

The platform combines **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **LangChain**, **LangGraph**, **AI Agents**, **SQLite**, **APIs**, and **AI Automation (n8n)** into one integrated system. It enables users to learn new technologies, build projects, assess their skills, receive personalized roadmaps, prepare for interviews, discover relevant jobs, and automate career-related tasks.

The goal is to provide an end-to-end AI-powered ecosystem that supports users from learning a skill to successfully applying for jobs.

2. Problem Statement

Students often use multiple disconnected platforms for learning, project development, interview preparation, job searching, and progress tracking.

For example:

- Learning from PDFs
- Watching YouTube tutorials
- Reading documentation
- Searching GitHub repositories
- Preparing resumes
- Applying for jobs
- Tracking interview progress

Because these resources are scattered across different platforms, learners struggle to organize their learning path and career development effectively.

Current AI chatbots answer questions but do not provide structured guidance, personalized learning, project recommendations, career tracking, or automation.

SkillForge AI addresses this challenge by bringing all these capabilities together into a single intelligent platform.

3. Project Objectives

The primary objectives of this project are:

- Build a Multi-Agent AI System using LangGraph.
 - Implement Retrieval-Augmented Generation (RAG) for document-based learning.
 - Integrate external APIs for real-world information.
 - Store user progress using SQLite.
 - Provide personalized learning roadmaps.
 - Recommend projects based on learned skills.
 - Analyze resumes and identify skill gaps.
 - Recommend relevant jobs.
 - Conduct AI-powered mock interviews.
 - Automate repetitive tasks using n8n.
 - Deliver an industry-inspired AI career platform.
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4. Proposed Solution

SkillForge AI functions as an AI Career Operating System.

Instead of acting as a single chatbot, it consists of multiple specialized AI agents coordinated by a Supervisor Agent.

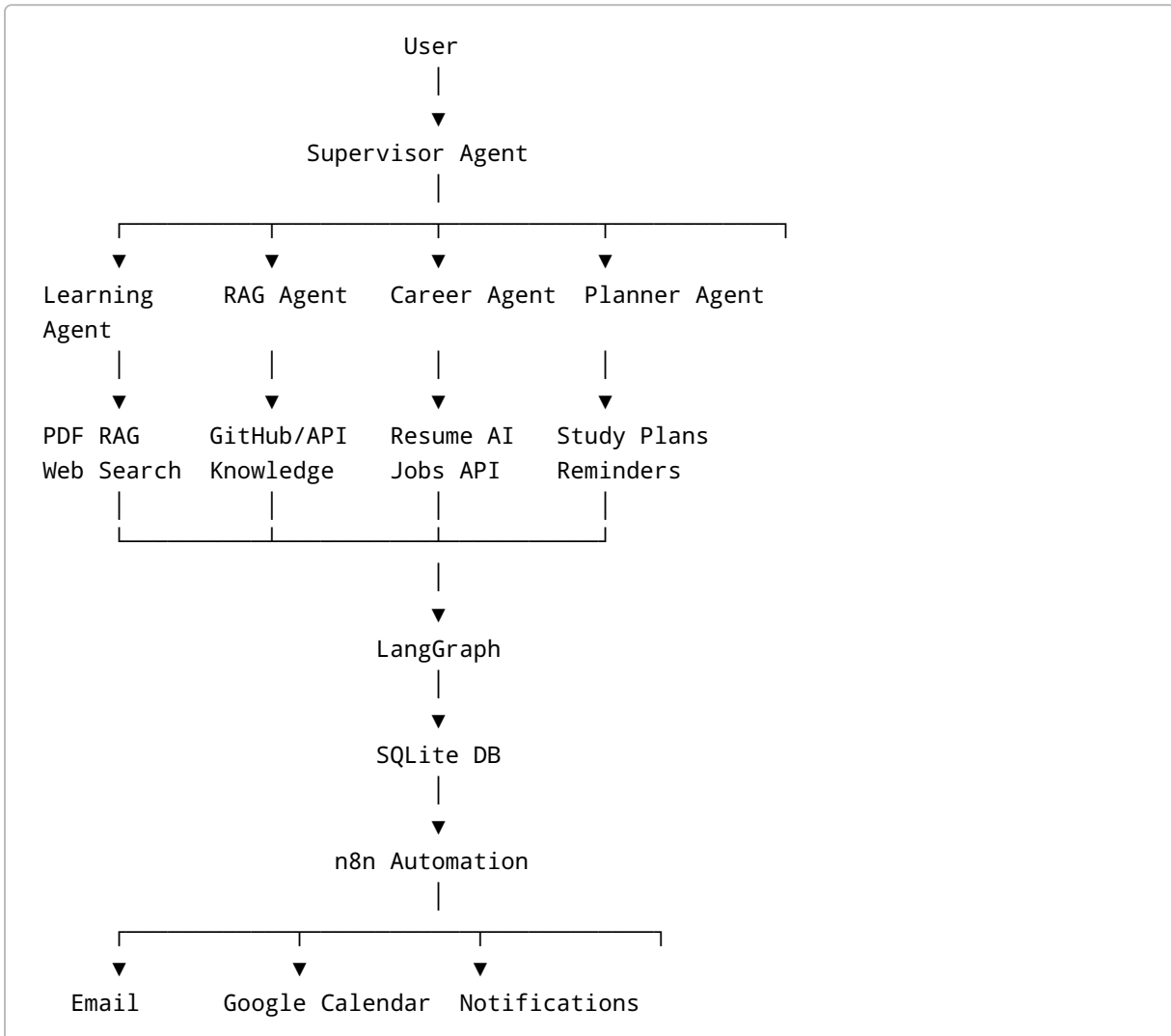
Each agent focuses on a specific responsibility.

Examples include:

- Learning Agent
- RAG Knowledge Agent
- Career Agent
- Resume Agent
- Interview Agent
- Planner Agent
- Automation Agent

The Supervisor Agent analyzes the user's request and delegates tasks to the appropriate specialized agents.

5. System Architecture



6. Major Modules

Module 1 – Learning Assistant

Features:

- Personalized Roadmaps
- Learning Recommendations
- Progress Tracking
- Weak Topic Detection

Technologies:

- LangChain
 - LLM
 - SQLite
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Module 2 – RAG Knowledge Assistant

Features:

- PDF Upload
- Semantic Search
- Context Retrieval
- AI Question Answering

Technologies:

- LangChain
 - ChromaDB
 - Embeddings
 - RAG
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Module 3 – Career Assistant

Features:

- Skill Assessment
- Job Recommendations
- Skill Gap Analysis
- Career Roadmaps

Technologies:

- APIs
 - LLM
 - SQLite
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Module 4 – Resume Assistant

Features:

- Resume Analysis
- Resume Improvement Suggestions

- Resume Tailoring for Job Descriptions

Technologies:

- LLM
 - Prompt Engineering
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Module 5 – Interview Assistant

Features:

- AI Mock Interviews
- Technical Questions
- Behavioral Questions
- Feedback and Scoring

Technologies:

- LangGraph
 - LLM
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Module 6 – Automation Assistant

Features:

- Email Reminders
- Weekly Progress Reports
- Calendar Scheduling
- Learning Notifications

Technologies:

- n8n
 - Gmail API
 - Google Calendar API
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7. Technology Stack

Category	Technology
Programming Language	Python
AI Framework	LangChain

Category	Technology
Agent Framework	LangGraph
LLM	Groq (Llama 3.3-70B) (<i>configurable</i>)
RAG	ChromaDB + Embeddings
Database	SQLite
Backend	FastAPI
Frontend	Streamlit
Automation	n8n
Version Control	Git & GitHub
Deployment	Docker (Future Phase)

8. Key Features

- Multi-Agent AI Architecture
- Supervisor Agent
- Retrieval-Augmented Generation (RAG)
- PDF Question Answering
- Personalized Learning Plans
- AI Career Guidance
- Resume Analysis
- Skill Gap Detection
- Job Recommendations
- AI Mock Interviews
- SQLite-Based User Memory
- Progress Dashboard
- AI Automation Workflows
- Email Notifications
- Calendar Integration

9. Expected Outcomes

Upon completion, SkillForge AI will:

- Help learners organize their learning journey.
- Answer questions using uploaded documents.
- Track learning progress.
- Recommend projects based on acquired skills.

- Identify missing skills required for target careers.
 - Recommend suitable job opportunities.
 - Prepare users for technical interviews.
 - Automate reminders and reports.
 - Demonstrate a complete Agentic AI workflow suitable for academic and professional portfolios.
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10. Future Enhancements

Future versions may include:

- Voice Assistant
 - Mobile Application
 - Authentication & User Accounts
 - Multi-user Support
 - MCP (Model Context Protocol)
 - Cloud Database Integration
 - Vector Database Scaling
 - Real-time Collaboration
 - AI Coding Assistant
 - AI Project Reviewer
 - Analytics Dashboard
 - SaaS Deployment
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11. Conclusion

SkillForge AI aims to move beyond the traditional chatbot model by delivering a comprehensive AI-powered career operating system. Through the integration of Multi-Agent AI, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, SQLite, APIs, and AI Automation, the platform provides an intelligent, personalized, and scalable solution for modern learners.

The project demonstrates advanced concepts in Agentic AI while solving a genuine real-world problem: guiding users from learning new skills to building projects, preparing for interviews, and progressing toward employment. It serves as both a practical educational platform and a strong portfolio project showcasing modern AI engineering practices.